Federated Multimodal Graph Learning Framework for Mortality and Sepsis Prediction using eICU

Objective

The objective is to design a federated learning framework that predicts **in-hospital mortality** and **sepsis** using multimodal patient data from the eICU dataset. Each hospital acts as an independent federated client, and patient data never leave the hospital. Only model parameters are shared with a central server.

Step 1: Data Preparation

For each hospital (identified using the `hospitalid` field in the eICU dataset):

- Collect all patients belonging to that hospital.
- Divide the patient data into three modalities:
- **Time-series data:** Vital signs, laboratory measurements, physiological signals collected over time.
- **Tabular data:** Demographics, admission information, diagnosis codes, severity scores, etc.
- **Unstructured data:** Clinical notes (if available) or other free-text information.

Each hospital performs all preprocessing locally.

Step 2: Modality-specific Feature Learning

Each modality is processed using a separate encoder.

Time-series Encoder

- Model: LSTM (or GRU/Transformer in future versions)
- Input: Sequential physiological measurements
- Output: Temporal embedding for each patient

Tabular Encoder

- Model: Autoencoder (or an MLP)
- Input: Structured clinical variables
- Output: Latent tabular embedding

Text Encoder

- Model: MiniBERT (or ClinicalBERT if available)
- Input: Clinical notes
- Output: Text embedding

Each encoder learns a compact representation of the patient's information.

Step 3: Multimodal Fusion

The embeddings from the three encoders are combined to obtain a unified patient representation.

Initially, simple concatenation can be used.

Patient Representation:

Time-series Embedding + Tabular Embedding + Text Embedding

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Unified Patient Embedding

(An attention-based or gated fusion module can later replace simple concatenation.)

Step 4: Intra-hospital Patient Graph Construction

Using the unified patient embeddings, construct a patient similarity graph within each hospital.

Possible approaches:

- k-Nearest Neighbour (k-NN)
- Cosine similarity with a predefined threshold

Each node represents one patient.

Edges connect patients with similar clinical characteristics.

No edges are created between patients from different hospitals.

Step 5: Graph Representation Learning

Train a Graph Neural Network (GraphSAGE, GAT, or GCN) on the patient graph.

Input:

- Patient graph
- Unified patient embeddings

Output:

- Graph-enhanced patient embeddings

The GNN captures relationships among clinically similar patients.

Step 6: Multi-task Prediction

Attach two prediction heads to the GNN output.

Task 1:

- Mortality Prediction

Task 2:

- Sepsis Prediction

The entire local model (encoders + fusion + GNN + prediction heads) is trained jointly using a multi-task loss.

Step 7: Federated Learning

Each hospital acts as one federated client.

The local workflow is:

1. Receive the current global model from the server.
2. Train the complete model using only local hospital data.
3. Update all model parameters.
4. Send only the updated model parameters (weights) to the central server.

Important: Raw patient data, embeddings, and graphs are never shared.

Step 8: Central Server

The server receives model parameters from all participating hospitals.

It performs federated aggregation (e.g., FedAvg initially) to produce an updated global model.

The updated global parameters are then sent back to every hospital for the next training round.

The server never has access to patient records.

Overall Workflow

For each hospital:

Patient Data

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Separate by Modality

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LSTM (Time-series)

Autoencoder (Tabular)

MiniBERT (Text)

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Fusion

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Unified Patient Embeddings

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Patient Similarity Graph

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Graph Neural Network

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Graph Embeddings

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Mortality Prediction

Sepsis Prediction

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Local Training

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Model Parameters

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Federated Server

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Parameter Aggregation

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Updated Global Model

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Back to All Hospitals

Deliverables

The implementation should include:

1. Data preprocessing pipeline.
2. Three modality-specific encoders.
3. Multimodal fusion module.
4. Patient graph construction algorithm.
5. Graph Neural Network implementation.
6. Multi-task prediction module.
7. Federated learning pipeline.
8. Training and evaluation scripts.
9. Baseline comparisons:
10. Local model only
11. Centralized model
12. Federated model (FedAvg)
13. Performance metrics:
14. AUROC

15. AUPRC
16. Accuracy
17. Precision
18. Recall
19. F1-score
20. Calibration (optional)

Future Improvements

After obtaining a working implementation, the following enhancements can be explored:

- Replace concatenation with attention-based multimodal fusion.
- Replace FedAvg with personalized or adaptive federated aggregation.
- Build dynamic temporal patient graphs instead of static similarity graphs.
- Introduce explainability using GNNExplainer or attention visualization.
- Incorporate uncertainty estimation and model calibration.